

Preethi Kurapaty

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Summary

Results-driven Data Analyst with over 3 years of experience in analyzing complex datasets, generating actionable insights, and supporting data-driven decisions. Proficient in Python, SQL, and data visualization tools to clean, interpret, and present data effectively. Skilled in applying statistical methods and predictive modeling to solve business problems and optimize processes. Collaborative team player with a strong focus on identifying trends, improving strategies, and enhancing operational efficiency while delivering high-quality, data-backed solutions.

Technical Skills

- **Programming Languages:** SQL, Python, R, Java
- **Databases:** MySQL, Snowflake, Redshift
- **Data Analysis Tools:** QuickSight, Spreadsheets, Google Analytics, JIRA, GitHub, Tableau, Power BI
- **Machine Learning & Statistical techniques:** Exploratory Data Analysis (EDA), A/B Testing, Data Wrangling, Statistical Modeling, Feature Engineering, Deep Learning, ETL, Business Process Improvement.
- **Libraries:** Pandas, Seaborn, Numpy, Keras, Matplotlib
- **Project Management & Methodologies:** Scrum, Agile
- **Certifications:** Databricks Certified Data Engineer Associate

Professional Experience

Graduate Teaching Assistant, Oregon State University 03/2023 – Present | Corvallis, USA

- Spearheaded the development and optimization of a robust real-time traffic monitoring system using Python and advanced frameworks such as Flask.
- Employed Geographic Information System (GIS) tools and predictive analytics to improve the accuracy and reliability of traffic data analysis.
- Conducted extensive testing and validation processes to ensure system performance and reliability under various traffic scenarios.
- Developed and implemented AI/ML models for classifying traffic patterns, leveraging Bayesian inference to enhance prediction accuracy.
- Assisted with the development of data transfer pipelines and storage solutions for large-scale transportation datasets using cloud computing platforms.
- Conducted research on applying machine learning techniques for echo classification and signal processing in real-time monitoring systems.

Software Development Engineer, Amazon 10/2022 – 03/2023 | Seattle, USA

- Worked on extracting and integrating CRM data into a centralized data warehouse using Snowflake and Azure Databricks, enabling better accessibility and analysis for business teams.
- Designed and built interactive Tableau dashboards to highlight customer dissatisfaction drivers and analyze behavioral patterns, providing actionable insights for improving customer experience.
- Utilized machine learning models, including Support Vector Machines (SVM), to predict customer churn and forecast satisfaction levels, helping the team proactively address churn risks.
- Conducted a detailed analysis of service quality, usage patterns, and customer feedback to pinpoint key churn factors and develop targeted retention strategies.
- Developed AI/ML-driven data pipelines and cloud storage guidelines for handling large-scale customer datasets in AWS.
- Assisted in implementing AI/ML-based classification models to detect anomalous customer behavior, improving fraud detection capabilities.

Graduate Teaching Assistant, Oregon State University 09/2021 – 09/2022 | Corvallis, USA

- Led the design and optimization of a real-time traffic monitoring system using Python and frameworks like Flask, ensuring scalability and responsiveness.
- Integrated GIS tools and predictive analytics to enhance the accuracy and reliability of traffic data analysis and forecasting.
- Performed comprehensive testing and validation under diverse traffic conditions to ensure system robustness, performance, and reliability.
- Conducted research on AI/ML methodologies for classifying real-time sensor data, contributing to the development of robust classification frameworks.
- Proposed and tested various AI/ML and Bayesian modeling approaches for real-time traffic and acoustic data classification.
- Worked on cloud implementation strategies for handling large-scale sensor data and optimized processing algorithms for efficiency.

Data Analyst, Technosoft Engineering 07/2019 – 08/2021 | India

- Designed a machine learning model utilizing eigenvalue clustering and classification techniques to detect fraudulent transactions.
- Developed standardized metrics that optimized query processes, reducing analysis time by 50%.
- Automated reporting workflows by integrating multiple data sources and generating dynamic interactive dashboards using D3.js and Tableau.
- Built predictive models to identify potential defects in automotive components by analyzing both structured and unstructured data, leading to significant cost savings on recalls.
- Applied NLP techniques such as text classification, clustering, and Word2Vec to extract key insights from warranty data.

- Created risk indicators to detect and flag transactional anomalies by implementing SQL stored procedures, custom scripts, and analytic reports.
- Established business rules, including triggers, check constraints, logging mechanisms, and relational integrity constraints in SQL Server, reducing update and delete errors by 40%.

Academic Projects

Time Series Analysis and Forecasting of Pharmacy Sales | Tools: SAS, Excel

- Forecasted sales of 8 categories of drugs in SAS EG for a pharmacy store based on 10 years of sales data.
- Performed trend and seasonality analysis on the dataset. ARIMA and ARIMAX models were used for forecasting and the **ARIMA model** had the lowest AIC value of **666.6**.

Breast cancer classification | Tools: Python, Tableau, Excel

- Evaluated Random Forest, Decision Tree, and Logistic Regression models using accuracy, precision, recall, and F1-score.
- Random Forest achieved **88.6%** accuracy, outperforming Decision Tree, and Logistic Regression by **8%**.
- Statistical testing confirmed strong generalization and a **20%** improvement in F1-score on unseen data.

Education

Master of Engineering in Computer Science, *Oregon State University*

Oregon, USA

Bachelor of Technology in Computer Science, *CVR College of Engineering*

Hyderabad, India